Ex.No:7(a) IMPLEMENT THE SYSTEM AS PER THE DETAILED DESIGN - M.SUGANYA - 8649

1. TECHNOLOGY STACK

- **Programming Language**: Java
- **Database**: MySQL
- **JDBC**: To connect Java with MySQL
- **IDE**: VScode
- **Database Tool**: MySQL Workbench

2. IMPLEMENTATION STEPS

A. Setup MySQL Database

1. Open **MySQL Workbench**
2. Create a database:

```
CREATE DATABASE exam_db;
```

3. Create a table for exam registration:

```
CREATE TABLE exam_registration (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_name VARCHAR(255),
    reg_no VARCHAR(50),
    subject VARCHAR(100),
    exam_date DATE
    );
```

B. Establish Database Connection Using JDBC

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
public class DBConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/exam_registration_db";
    private static final String USER = "root";
    private static final String PASSWORD = " ";
    public static Connection getConnection() {
        Connection connection = null;
```

```
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            connection = DriverManager.getConnection(URL, USER, PASSWORD);
        } catch (ClassNotFoundException | SQLException e) {
            e.printStackTrace();
        }
        return connection;
    }
}
```

C. Implement the Operations

1. Create a Model Class

```
public class ExamRegistration {
    private int id;
    private String studentName;
    private String registerNumber;
    private String subject;
    private String examDate;

    public ExamRegistration(int id, String studentName, String registerNumber, String subject,
String examDate) {
        this.id = id;
        this.studentName = studentName;
        this.registerNumber = registerNumber;
        this.subject = subject;
        this.examDate = examDate;
    }
    public int getId() { return id; }
    public String getStudentName() { return studentName; }
    public String getRegisterNumber() { return registerNumber; }
    public String getSubject() { return subject; }
    public String getExamDate() { return examDate; }
```

```
}
```

2. Create DAO Class

```
import java.sql.*;
import java.util.*;
public class ExamDAO {
    private Connection connection;
    public ExamDAO() {
        this.connection = DBConnection.getConnection();
    }
    public void addRegistration(ExamRegistration reg) {
        String sql = "INSERT INTO exam_registration (student_name, register_number, subject,
exam_date) VALUES (?, ?, ?, ?)";
        try (PreparedStatement stmt = connection.prepareStatement(sql)) {
            stmt.setString(1, reg.getStudentName());
            stmt.setString(2, reg.getRegisterNumber());
            stmt.setString(3, reg.getSubject());
            stmt.setString(4, reg.getExamDate());
            stmt.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
    public List<ExamRegistration> getAllRegistrations() {
        List<ExamRegistration> list = new ArrayList<>();
        String sql = "SELECT * FROM exam_registration";
        try (Statement stmt = connection.createStatement();
             ResultSet rs = stmt.executeQuery(sql)) {
            while (rs.next()) {
                list.add(new ExamRegistration(
                        rs.getInt("id"),
                        rs.getString("student_name"),
```

```
                rs.getString("register_number"),
                rs.getString("subject"),
                rs.getString("exam_date")
            ));
        }
    } catch (SQLException e) {
        e.printStackTrace();
    }
    return list;
}
public boolean deleteRegistration(int id) {
    String query = "DELETE FROM registrations WHERE id = ?";
    try (Connection conn = DBConnection.getConnection();
         PreparedStatement stmt = conn.prepareStatement(query)) {
        stmt.setInt(1, id);
        int rowsAffected = stmt.executeUpdate();
        return rowsAffected > 0;
    } catch (SQLException e) {
        e.printStackTrace();
        return false;
    }
}
}
```

D. Create Main Application

```
import java.util.*;
public class ExamRegistrationApp {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        ExamDAO dao = new ExamDAO();
     try{
```

```
        while (true) {
            System.out.println("\nExam Registration System");
            System.out.println("1. Add Registration");
            System.out.println("2. View Registrations");
            System.out.println("3. Delete Registration");
            System.out.println("4. Exit");
            System.out.print("Enter choice: ");
            int choice = scanner.nextInt();
            scanner.nextLine();
            switch (choice) {
                case 1:
                    System.out.print("Student Name: ");
                    String name = scanner.nextLine();
                    System.out.print("regis no: ");
                    String regNo = scanner.nextLine();
                    System.out.print("Subject: ");
                    String subject = scanner.nextLine();
                    System.out.print("Exam Date (YYYY-MM-DD): ");
                    String date = scanner.nextLine();
                    ExamRegistration reg = new ExamRegistration(0, name, regNo, subject, date);
                    dao.addRegistration(reg);
                    System.out.println("Registration added.");
                    break;
                case 2:
                    List<ExamRegistration> list = dao.getAllRegistrations();
                    System.out.println("\nRegistered Students:");
                    for (ExamRegistration r : list) {
                        System.out.println(r.getId() + ": " + r.getStudentName() + " - " + r.getSubject()
+ " on " + r.getExamDate());
                    }
                    break;
```

```
            case 3:
                System.out.print("Enter ID to delete: ");
                int id = scanner.nextInt();
                dao.deleteRegistration(id);
                System.out.println("Registration deleted.");
                break;
            case 4:
                System.out.println("Goodbye!");
                System.exit(0);
            default:
                System.out.println("Invalid option.");
        }
    }
}
    finally {
        scanner.close();
    }
}
}
```

OUTPUT:

```
PS D:\문서\sem 6\oose\practice\PRACTICE> java -cp "bin;lib/mysql-connector-j-9.3.0.

Exam Registration System
1. Add Registration
2. View Registrations
3. Delete Registration
4. Exit
Enter choice: 2

Registered Students:
1: Sugan - Math on 2025-06-15
2: Mithra - Physics on 2025-06-20
3: Valar - Chemistry on 2025-06-22
4: Bob - Biology on 2025-06-25
5: su - cs on 2023-12-12

Exam Registration System
1. Add Registration
2. View Registrations
3. Delete Registration
4. Exit
Enter choice: 1
Student Name: Luca
regis no: 123445
Subject: ds
Exam Date (YYYY-MM-DD): 2021-12-23
Registration added.

Exam Registration System
1. Add Registration
2. View Registrations
3. Delete Registration
```

Ex.No:7(c) IMPLEMENT THE SYSTEM AS PER THE DETAILED DESIGN – M.SUGANYA - 8649

1. TECHNOLOGY STACK

- **Programming Language**: Java
- **Database**: MySQL
- **JDBC**: To connect Java with MySQL
- **IDE**: VScode
- **Database Tool**: MySQL Workbench

2. IMPLEMENTATION STEPS

A. Setup MySQL Database

1. Open **MySQL Workbench**
2. Create a database:

```
CREATE DATABASE student_info_db;
```

3. Create a table for exam registration:

```
CREATE TABLE students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    age INT,
    dob DATE,
    department VARCHAR(100),
    enrollment_year YEAR
);
```

B. Establish Database Connection Using JDBC

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
public class DBConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/student_info_db";
    private static final String USER = "root";
    private static final String PASSWORD = "";
```

```
    public static Connection getConnection() {
        Connection connection = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            connection = DriverManager.getConnection(URL, USER, PASSWORD);
        } catch (ClassNotFoundException e) {
            System.err.println("⚠ JDBC Driver not found!");
            e.printStackTrace();
        } catch (SQLException e) {
            System.err.println("Failed to connect to MySQL!");
            e.printStackTrace();
        }
        return connection;
    }
}
```

C. Implement the Operations

1. Create a Model Class

```
public class Student {
    private int id;
    private String name;
    private String email;
    private int age;
    private String dob;
    private String department;
    private int enrollmentYear;
    public Student(int id, String name, String email, int age, String dob, String department, int
enrollmentYear) {
        this.id = id;
        this.name = name;
        this.email = email;
        this.age = age;
```

```
        this.dob = dob;
        this.department = department;
        this.enrollmentYear = enrollmentYear;
    }
    public int getId() { return id; }
    public String getName() { return name; }
    public String getEmail() { return email; }
    public int getAge() { return age; }
    public String getDob() { return dob; }
    public String getDepartment() { return department; }
    public int getEnrollmentYear() { return enrollmentYear; }
}
```

2. Create DAO Class

```
import java.sql.*;
import java.util.ArrayList;
import java.util.List;
public class StudentDAO {
    private Connection connection;
    public StudentDAO() {
        this.connection = DBConnection.getConnection();
        if (this.connection == null) {
            throw new IllegalStateException("Database connection failed! Check your
DBConnection class.");
        }
    }
    public void addStudent(Student student) {
        String sql = "INSERT INTO students (name, email, age, dob, department,
enrollment_year) VALUES (?, ?, ?, ?, ?, ?)";
        try (PreparedStatement stmt = connection.prepareStatement(sql)) {
            stmt.setString(1, student.getName());
            stmt.setString(2, student.getEmail());
```

```
        stmt.setInt(3, student.getAge());
        stmt.setString(4, student.getDob());
        stmt.setString(5, student.getDepartment());
        stmt.setInt(6, student.getEnrollmentYear());
        stmt.executeUpdate();
    } catch (SQLException e) {
        System.err.println(" Error executing INSERT query!");
        e.printStackTrace();
    }
}
public List<Student> getAllStudents() {
    List<Student> students = new ArrayList<>();
    String sql = "SELECT * FROM students";
    try (Statement stmt = connection.createStatement();
         ResultSet rs = stmt.executeQuery(sql)) {
        while (rs.next()) {
            Student student = new Student(
                rs.getInt("id"),
                rs.getString("name"),
                rs.getString("email"),
                rs.getInt("age"),
                rs.getString("dob"),
                rs.getString("department"),
                rs.getInt("enrollment_year")
            );
            students.add(student);
        }
    } catch (SQLException e) {
        System.err.println(" Error retrieving students!");
        e.printStackTrace();
```

```
        }
        return students;
    }
    public void deleteStudent(int studentId) {
        String sql = "DELETE FROM students WHERE id = ?";
        try (PreparedStatement stmt = connection.prepareStatement(sql)) {
            stmt.setInt(1, studentId);
            stmt.executeUpdate();
        } catch (SQLException e) {
            System.err.println("Error deleting student!");
            e.printStackTrace();
        }
    }
}
```

D. Create Main Application

```
import java.util.List;
import java.util.Scanner;
public class StudentInfoApp {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        StudentDAO studentDAO = new StudentDAO();
        try {
            while (true) {
                System.out.println("\n Student Information System");
                System.out.println("1. Add a Student");
                System.out.println("2. View Students");
                System.out.println("3. Delete a Student");
                System.out.println("4. Exit");
                System.out.print("Enter choice: ");
                int choice = scanner.nextInt();
```

```
            scanner.nextLine();
            switch (choice) {
                case 1:
                    System.out.print("Enter Name: ");
                    String name = scanner.nextLine();
                    System.out.print("Enter Email: ");
                    String email = scanner.nextLine();
                    System.out.print("Enter Age: ");
                    int age = scanner.nextInt();
                    scanner.nextLine();
                    System.out.print("Enter Date of Birth (YYYY-MM-DD): ");
                    String dob = scanner.nextLine();
                    System.out.print("Enter Department: ");
                    String department = scanner.nextLine();
                    System.out.print("Enter Enrollment Year: ");
                    int enrollmentYear = scanner.nextInt();
                    Student student = new Student(0, name, email, age, dob, department,
enrollmentYear);
                    studentDAO.addStudent(student);
                    System.out.println("Student added successfully!");
                    break;
                case 2:
                    List<Student> students = studentDAO.getAllStudents();
                    System.out.println("\n Registered Students:");
                    for (Student s : students) {
                        System.out.println(s.getId() + ". " + s.getName() + " | " + s.getEmail() + " |
Age: " + s.getAge() + " | DOB: " + s.getDob() + " | Dept: " + s.getDepartment() + " |
Enrollment Year: " + s.getEnrollmentYear());
                    }
                    break;
                case 3:
```

```
                System.out.print("Enter Student ID to delete: ");
                int studentId = scanner.nextInt();
                studentDAO.deleteStudent(studentId);
                System.out.println(" Student deleted successfully!");
                break;
            case 4:
                System.out.println(" Exiting...");
                return;
            default:
                System.out.println(" Invalid choice! Try again.");
        }
    }
} finally {
    scanner.close();
}
}
}
```

OUTPUT:

```
PS D:\문서\sem 6\oose\practice\PRACTICE> java -cp "bin;lib/mysql-connector-j-9.3.0.jar" ExamRegistrationApp

Exam Registration System
1. Add Registration
2. View Registrations
3. Delete Registration
4. Exit
Enter choice: 2

Registered Students:
1: Sugan - Math on 2025-06-15
2: Mithra - Physics on 2025-06-20
3: Valar - Chemistry on 2025-06-22
4: Bob - Biology on 2025-06-25
5: su - cs on 2023-12-12

Exam Registration System
1. Add Registration
2. View Registrations
3. Delete Registration
4. Exit
Enter choice: 1
Student Name: Luca
regis no: 123445
Subject: ds
Exam Date (YYYY-MM-DD): 2021-12-23
Registration added.

Exam Registration System
1. Add Registration
2. View Registrations
3. Delete Registration
```

Ex.No:7(b) IMPLEMENT THE SYSTEM AS PER THE DETAILED DESIGN – M.SUGANYA - 8649

1. TECHNOLOGY STACK

- **Programming Language**: Java
- **Database**: MySQL
- **JDBC**: To connect Java with MySQL
- **IDE**: VScode
- **Database Tool**: MySQL Workbench

2. IMPLEMENTATION STEPS

A. Setup MySQL Database

4. Open **MySQL Workbench**
5. Create a database:

```
CREATE DATABASE book_bank_db;
```

4. Create a table for exam registration:

```
CREATE TABLE books (
    id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    author VARCHAR(100),
    isbn VARCHAR(20) UNIQUE,
    available BOOLEAN DEFAULT TRUE
);
```

B. Establish Database Connection Using JDBC

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/book_bank_db";
    private static final String USER = "root";
    private static final String PASSWORD = "";
```

```
    public static Connection getConnection() {
        Connection connection = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            connection = DriverManager.getConnection(URL, USER, PASSWORD);
        } catch (ClassNotFoundException e) {
            System.err.println("JDBC Driver not found!");
            e.printStackTrace();
        } catch (SQLException e) {
            System.err.println("Failed to connect to MySQL!");
            e.printStackTrace();
        }
        return connection;
    }
}
```

C. Implement the Operations

1. Create a Model Class

```
public class Book {
    private int id;
    private String title;
    private String author;
    private String isbn;
    private boolean available;
    public Book(int id, String title, String author, String isbn, boolean available) {
        this.id = id;
        this.title = title;
        this.author = author;
        this.isbn = isbn;
        this.available = available;
    }
```

```
    public int getId() { return id; }
    public String getTitle() { return title; }
    public String getAuthor() { return author; }
    public String getIsbn() { return isbn; }
    public boolean isAvailable() { return available; }
}
```

2. Create DAO Class

```
import java.sql.*;
import java.util.ArrayList;
import java.util.List;
public class BookDAO {
    private Connection connection;
    public BookDAO() {
        this.connection = DBConnection.getConnection();
        if (this.connection == null) {
            throw new IllegalStateException("Database connection failed! Check your
DBConnection class.");
        }
    }
    public void addBook(Book book) {
        String sql = "INSERT INTO books (title, author, isbn, available) VALUES (?, ?, ?, ?)";
        try (PreparedStatement stmt = connection.prepareStatement(sql)) {
            stmt.setString(1, book.getTitle());
            stmt.setString(2, book.getAuthor());
            stmt.setString(3, book.getIsbn());
            stmt.setBoolean(4, book.isAvailable());
            stmt.executeUpdate();
        } catch (SQLException e) {
            System.err.println(" Error executing INSERT query!");
            e.printStackTrace();
```

```
    }
}
public List<Book> getAllBooks() {
    List<Book> books = new ArrayList<>();
    String sql = "SELECT * FROM books";
    try (Statement stmt = connection.createStatement();
         ResultSet rs = stmt.executeQuery(sql)) {
        while (rs.next()) {
            Book book = new Book(
                rs.getInt("id"),
                rs.getString("title"),
                rs.getString("author"),
                rs.getString("isbn"),
                rs.getBoolean("available")
            );
            books.add(book);
        }
    } catch (SQLException e) {
        System.err.println(" Error retrieving books!");
        e.printStackTrace();
    }
    return books;
}
public void deleteBook(int bookId) {
    String sql = "DELETE FROM books WHERE id = ?";
    try (PreparedStatement stmt = connection.prepareStatement(sql)) {
        stmt.setInt(1, bookId);
        stmt.executeUpdate();
    } catch (SQLException e) {
        System.err.println(" Error deleting book!");
```

```
            e.printStackTrace();
        }
    }
}
```

D. Create Main Application

```
import java.util.List;
import java.util.Scanner;
public class BookBankApp {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        BookDAO bookDAO = new BookDAO();
        try {
            while (true) {
                System.out.println("\n Book Bank Management System");
                System.out.println("1. Add a Book");
                System.out.println("2. View Books");
                System.out.println("3. Delete a Book");
                System.out.println("4. Exit");
                System.out.print("Enter choice: ");
                int choice = scanner.nextInt();
                scanner.nextLine();
                switch (choice) {
                    case 1:
                        System.out.print("Enter Title: ");
                        String title = scanner.nextLine();
                        System.out.print("Enter Author: ");
                        String author = scanner.nextLine();
                        System.out.print("Enter ISBN: ");
                        String isbn = scanner.nextLine();
```

```
                Book book = new Book(0, title, author, isbn, true);
                bookDAO.addBook(book);
                System.out.println(" Book added successfully!");
                break;
            case 2:
                List<Book> books = bookDAO.getAllBooks();
                System.out.println("\n Books in Book Bank:");
                for (Book b : books) {
                    System.out.println(b.getId() + ". " + b.getTitle() + " by " + b.getAuthor());
                }
                break;
            case 3:
                System.out.print("Enter Book ID to delete: ");
                int bookId = scanner.nextInt();
                bookDAO.deleteBook(bookId);
                System.out.println(" Book deleted successfully!");
                break;
            case 4:
                System.out.println("Exiting...");
                return;
            default:
                System.out.println("Invalid choice! Try again.");
            }
        }
    } finally {
        scanner.close();
    }
  }
}
```

OUTPUT:

```
? Book Bank Management System
1. Add a Book
2. View Books
3. Delete a Book
4. Exit
Enter choice: 2

? Books in Book Bank:
1. The Great Gatsby by F. Scott Fitzgerald
2. To Kill a Mockingbird by Harper Lee
3. 1984 by George Orwell
4. The Catcher in the Rye by J.D. Salinger

? Book Bank Management System
1. Add a Book
2. View Books
3. Delete a Book
4. Exit
Enter choice: 1
Enter Title: The scar
Enter Author: Sugan
Enter ISBN: 658888324545
? Book added successfully!

? Book Bank Management System
1. Add a Book
2. View Books
3. Delete a Book
4. Exit
Enter choice: 2

? Books in Book Bank:
1. The Great Gatsby by F. Scott Fitzgerald
2. To Kill a Mockingbird by Harper Lee
3. 1984 by George Orwell
4. The Catcher in the Rye by J.D. Salinger
5. The scar by Sugan

? Book Bank Management System
1. Add a Book
2. View Books
```